

2. Benchmarks

2.1 The standardised suite: HGB

Heterogeneous GNN papers before 2021 each preprocessed DBLP/ACM/IMDB their own way, so their reported numbers were not comparable. **HGB** (Heterogeneous Graph Benchmark; Lv, Ding, Liu et al., KDD 2021) exists precisely to fix this: fixed splits, fixed features, one leaderboard. Its headline statistics are Table ??; the per-type breakdown is Table ??.

Table 1:

Task	Dataset	#Nodes	#N. types	#Edges	#E. types	Target	#Classes
NC	DBLP	26,128	4	239,566	6	author	4
	IMDB	21,420	4	86,642	6	movie	5 [†]
	ACM	10,942	4	547,872	8	paper	3
	Freebase	180,098	8	1,057,688	36	book	7
LP	Amazon	10,099	1	148,659	2	product-product	—
	LastFM	20,612	3	141,521	3	user-artist	—
	PubMed	63,109	4	244,986	10	disease-disease	—

[†] **multi-label** — see Section ??.

Table 2:

Dataset	Node type	Count	Feat. dim	Relation	#Edges
DBLP	author *	4,057	334	author-paper	19,645
	paper	14,328	4,231	paper-term	85,810
	term	7,723	50	paper-venue	14,328
	venue	20	—	(+3 inverse relations)	
IMDB	movie *	4,932	3,489	movie-director	4,932
	director	2,393	3,341	movie-actor	14,779
	actor	6,124	3,341	movie-keyword	23,610
	keyword	7,971	—	(+3 inverse relations)	
ACM	paper *	3,025	1,902	paper-cite-paper	5,343
	author	5,959	1,902	paper-ref-paper	5,343
	subject	56	1,902	paper-author	9,949
	term	1,902	—	paper-subject	3,025
				paper-term	255,619
Freebase	book *	40,402	film 19,427	music 82,351	
	sports	1,025	people 17,641	location 9,368	
	organization	2,731	business 7,153	no features at all	
	36 typed relations (BOOK-AND-BOOK 202,674, MUSIC-AND-MUSIC 283,670, ...)				

* labelled target type. Labelled counts: DBLP 4,057/4,057 authors; ACM 3,025/3,025 papers; IMDB 4,573/4,932 movies (of which **2,684 carry more than one label**); Freebase 7,954/40,402 books. Split is 24% train / 6% validation / 70% test on the labelled target nodes.

Provenance of Table ??. The HGB paper publishes only the totals of Table ??; it never prints a per-type breakdown. The numbers above were therefore recomputed directly from the released `node.dat/link.dat/label.dat` files, and every one of the four datasets reproduces its published

totals exactly (nodes, edges, number of node types, number of edge types and number of classes all match). That agreement is what licenses the table. Two incidental findings: HGB’s own `info.dat` declares the DBLP author attribute dimension as 0, but the data file actually carries 334-dimensional author features — and 334 is exactly the feature dimension HAN publishes for DBLP, which independently corroborates the data over the metadata; and ACM’s 1,902-dimensional bag-of-words is indexed by exactly the 1,902 `term` nodes.

2.2 The individual benchmark graphs

DBLP (bibliographic). A computer-science bibliography extract with types *author*, *paper*, *term*, *venue/conference*. Authors are labelled with one of 4 research areas (database, data mining, AI, information retrieval); author features are a bag-of-words over the keywords of the papers they wrote. It is the canonical author-classification HIN and the standard testbed for APA vs. APCPA semantics.

ACM (bibliographic). Papers from ACM venues with types *paper*, *author*, *subject*, *term*. Papers are labelled with 3 classes (database, wireless communication, data mining) and carry a bag-of-words feature. Distinctively, ACM has a *paper–paper* citation relation in addition to the star structure, and HGB splits it into a citation and a reference direction — so the schema is not a pure star.

IMDB (movie). Types *movie*, *director*, *actor* and (in HGB) *keyword*. Movies are labelled by genre. This is the benchmark where versions diverge most sharply: MAGNN’s IMDB has 3 *disjoint* genres (Action/Comedy/Drama), whereas HGB’s has 5 *overlapping* genres, making it multi-label.

Freebase. *The name denotes two unrelated graphs.* HGB’s Freebase is a book-centric knowledge graph: 8 node types, 36 relations, 180,098 nodes, target type BOOK with 7 classes, and **no node features whatsoever** — so it isolates how much a model can extract from pure typed structure. HeCo’s “Freebase” is instead a *movie* dataset (3,492 movies, 33,401 actors, 2,502 directors, 4,459 writers; target movie, 3 classes). Quoting a Freebase number without saying which one is meaningless.

AMiner (bibliographic). Two scales. metapath2vec’s AMiner is web-scale: 1,693,531 authors, 3,194,405 papers, 3,883 venues, with 8 Google-Scholar categories as labels and 246,678 labelled authors. HeCo’s AMiner is a small subset (6,564 papers, 13,329 authors, 35,890 references; target paper, 4 classes). *Erratum worth knowing:* metapath2vec’s body text says the dataset “consists of 9,323,739 computer scientists”, but the authors’ own correction note states that 9,323,739 is the paper–author *link* count and the author count is 1,693,531. The wrong figure is still widely re-quoted.

Yelp and Douban (business / review). HERec’s Yelp has types *user* (16,239), *business* (14,284), *city* (47), *category* (511) and *compliment* (11), with U–B 198,397, U–U 158,590, U–Co 76,875, B–Ci 14,267 and B–Ca 40,009 edges (density 0.08%). Douban Movie has *user* (13,367), *movie* (12,677), *director* (2,449), *actor* (6,311) and *type* (38). Both are used for rating prediction / recommendation, reported with MAE and RMSE rather than F1.

LastFM and MovieLens (user–item HINs). MAGNN’s Last.fm has *user* (1,892), *artist* (17,632) and *tag* (1,088), with U–U 12,717, U–A 92,834 and A–T 23,253; it carries *no labels and no features*, so it is a pure link-prediction benchmark (predict user–artist), with one-hot IDs as dummy inputs. MovieLens/ML-100K is used analogously as a user–item bipartite HIN enriched with occupation/genre types. **PubMed**, in its HGB heterogeneous form, is a 4-type biomedical graph (63,109 nodes, 10 relation types) used for disease–disease link prediction — not to be confused with the homogeneous PubMed citation network of Problem 1.

OGB: ogbn-mag and MAG240M. These supply the scale the older benchmarks lack. **ogbn-mag** is a Microsoft Academic Graph subset with *paper* (736,389), *author* (1,134,649), *institution* (8,740) and *field of study* (59,965) — 1,939,743 nodes — and four relations: author–affiliated_with–institution (1,043,998), author–writes–paper (7,145,660), paper–cites–paper (5,416,271) and paper–has_topic–field_of_study (7,505,078), totalling 21,111,007 edges. The task is predicting each paper’s **venue** among **349 classes**; only *paper* nodes have features (128-d word2vec), the rest are featureless. Crucially the split is **temporal** — train ≤ 2017 , validation 2018, test ≥ 2019 — not random, so it measures generalisation forward in time. The official metric is accuracy. **MAG240M** (OGB-LSC) is the same design two orders of magnitude larger: 121,751,666 papers, 122,383,112 authors and 25,721 institutions (244,160,499 nodes) with 1,728,364,232 edges, 768-d RoBERTa features, and 153 arXiv subject classes.

2.3 Version incompatibility — the caveat that matters

The same dataset name denotes different graphs in different papers. Table ?? makes this concrete; quoting a statistic without naming the version is an error.

Table 3:

Dataset	Version	Node counts	Note
DBLP	HAN	paper 14,328 / 14,327, author 4,057, term 8,789 , venue 20	P–T = 88,420
	MAGNN / HeCo / HGB	paper 14,328, author 4,057, term 7,723 , venue 20	P–T = 85,810
	GTN	18,405 total (= paper + author + venue; <i>term dropped</i>)	4 edge types
ACM	HAN	paper 3,025, author 5,835, subject 56	P–A = 9,744
	HGB	paper 3,025, author 5,959 , subject 56, term 1,902	P–A = 9,949
	HeCo	paper 4,019 , author 7,167 , subject 60	P–A = 13,407
IMDB	HAN	movie 4,780, director 2,269, actor 5,841	
	MAGNN	movie 4,278, director 2,081, actor 5,257	3 classes, single-label
	HGB	movie 4,932, director 2,393, actor 6,124, keyword 7,971	5 classes, multi-label

Four specific discrepancies are worth flagging.

- DBLP’s term count.** HAN reports 8,789 terms; MAGNN, HeCo and HGB all use 7,723. HGB’s published total is only consistent with the latter ($4,057 + 14,328 + 7,723 + 20 = 26,128$, whereas the HAN figure would give 27,194). HAN’s own table is also internally inconsistent: it lists 14,328 papers in the Paper–Author and Paper–Conference rows but 14,327 in the Paper–Term row.
- HGB’s ACM does not match its stated provenance.** HGB says it uses HAN’s subset while “preserving all edges including paper citations and references”, but its author count

(5,959 vs. 5,835) and P–A edge count (9,949 vs. 9,744) differ from HAN’s published table. Re-adding citation edges cannot change the number of *authors*, so the two ACM graphs genuinely differ.

3. **IMDB changes task type, not just size.** HGB’s IMDB is multi-label with 5 overlapping genres; MAGNN’s is single-label with 3 disjoint ones. An F1 on one is not comparable to an F1 on the other even in principle (Section ??).
4. **HGB’s LastFM edge count double-counts one relation.** Its node total 20,612 exactly equals MAGNN’s $1,892 + 17,632 + 1,088$, but its edge total is 141,521 against MAGNN’s 128,804. The gap is exactly 12,717, the U–U count: $2 \times 12,717 + 92,834 + 23,253 = 141,521$. So the symmetric user–user relation is stored in both directions while U–A and A–T are stored in one.